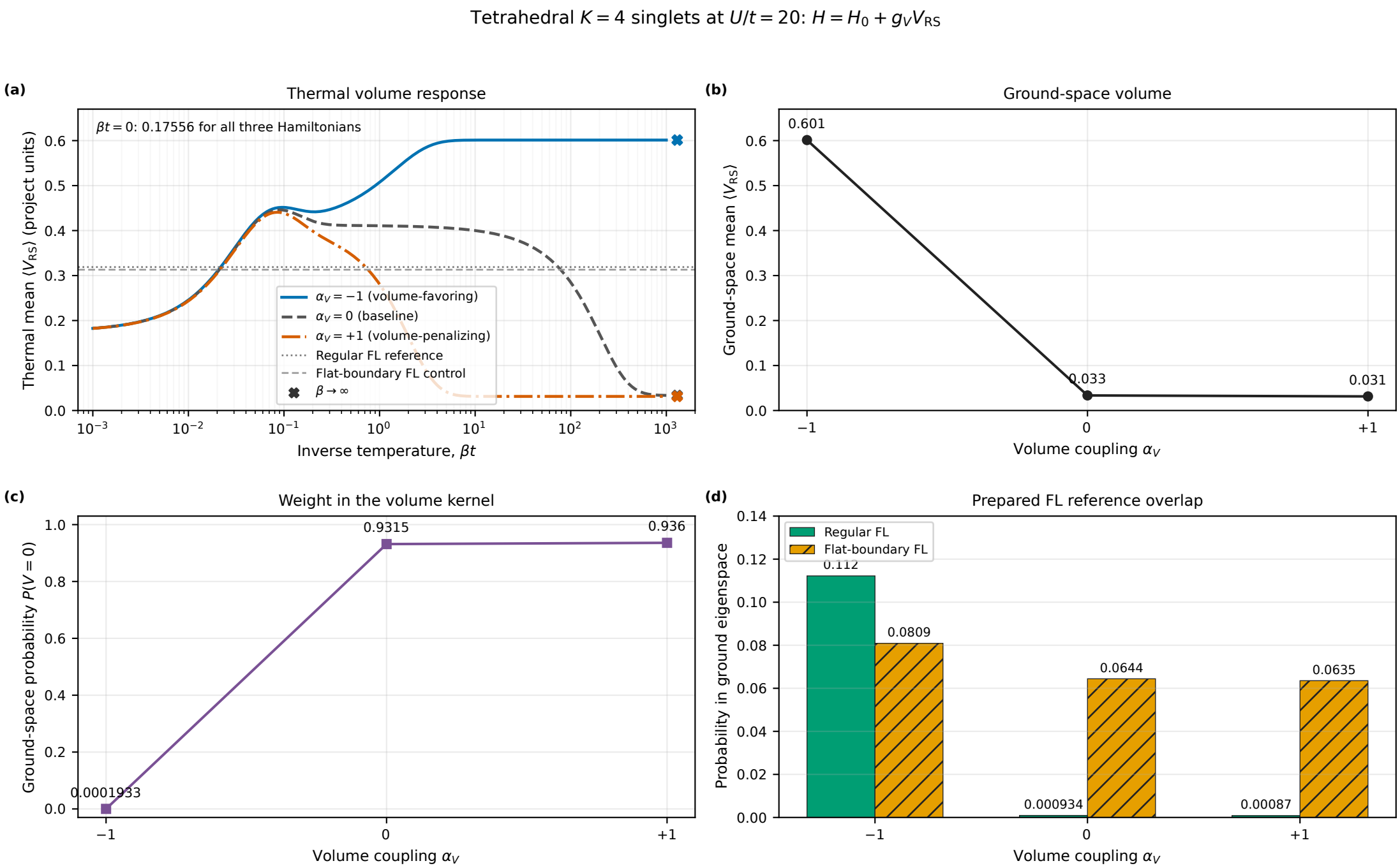
Tetrahedral $K = 4$ singlets at $U/t = 20$: $H = H_0 + g_V V_{RS}$
(a)
Thermal volume response
$\beta t = 0$: 0.17556 for all three Hamiltonians
Thermal mean $\langle V_{RS} \rangle$ (project units)
Inverse temperature, $\beta t$
$\alpha_V = -1$ (volume-favoring)
$\alpha_V = 0$ (baseline)
$\alpha_V = +1$ (volume-penalizing)
Regular FL reference
Flat-boundary FL control
$\beta \to \infty$
(b)
Ground-space volume
Ground-space mean $\langle V_{RS} \rangle$
Volume coupling $\alpha_V$
0.601
0.033
0.031
(c)
Weight in the volume kernel
Ground-space probability $P(V = 0)$
Volume coupling $\alpha_V$
0.0001933
0.9315
0.936
(d)
Prepared FL reference overlap
Probability in ground eigenspace
Volume coupling $\alpha_V$
Regular FL
Flat-boundary FL
0.112
0.0809
0.000934
0.0644
0.00087
0.0635